DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAMINATION

January 2025

Probability (Ph.D. Version)

Instructions to the Student

- Answer all six questions. Each will be graded from 0 to 10.
- Use a different booklet for each question. Write the problem number and your code number (NOT YOUR NAME) on the outside cover.
- Keep scratch work on separate pages in the same booklet.
- If you use a "well known" theorem in your solution to any problem, it is your responsibility to make clear which theorem you are using and to justify its use.

(1) Let $X \sim N(0,1)$ and Y be a random variable that is independent of X such that $\mathbb{P}(Y = 1) = \mathbb{P}(Y = -1) = \frac{1}{2}$. Show that (i) $YX \sim N(0,1)$, (ii) (X, XY) is NOT a Gaussian vector.

(2) Consider the probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ given by $\mu = \frac{1}{2}\mu_1 + \frac{1}{2}\mu_2$ with $\mu_1(\{1\}) = 1$ and $\mu_2((-\infty, x]) = \int_{-\infty}^{x} e^{-y} 1_{y \geq 0} dy$ for any $x \in \mathbb{R}$. Define the random variable $\xi : \mathbb{R} \to \mathbb{R}$ by $\xi(x) = x$, compute $\mathbb{E}\xi$ and $\mathbb{E}\xi^2$.

(3) Consider a symmetric simple random walk on $\{1, 2, 3, 4\}$ with reflecting boundary conditions, i.e., when the walker is at 2 or 3, it moves to one of its neighbors with probability $1/2$, and if the walker is at 1, it can only move to 2; when the walker is at 4, it can only move 3.

(i) Write down the transition matrix for the corresponding Markov chain.

(ii) Is there an invariant measure for the Markov chain? Is it unique? If so, find the invariant measure.

(iii) Is it true that the distribution of the chain at time n converges, as $n \to \infty$, for each initial distribution?

(4) Let $\{X_n\}_{n\geq 1}$ be a sequence of independent random variables such that

$$\mathbb{P}(X_n = 1) = \mathbb{P}(X_n = -1) = \frac{1}{2} - \frac{1}{2n^4}$$
$$\mathbb{P}(X_n = n^2) = \mathbb{P}(X_n = -n^2) = \frac{1}{2n^4}$$

(i) show that $\{X_n\}_{n\geq 1}$ does not satisfy the Lindeberg condition;

(ii) define another sequence of random variables $\{\tilde{X}_n\}_{n\geq 1}$ such that $\tilde{X}_n = 1$ if $X_n = 1, n^2$ and $\tilde{X}_n = -1$ if $X_n = -1, -n^2$, show that $\sum_{j=1}^n X_j - \sum_{j=1}^n \tilde{X}_j$ converges almost surely;

(iii) show that $\frac{X_1 + \ldots X_n}{\sqrt{n}}$ converges in distribution to a Gaussian random variable, and compute the mean and variance of the limit.

(5) Consider a standard Brownian motion B starting at $B_0 = 0$, let τ be the exit time of the Brownian motion from the interval $[-1, 1]$, i.e., the first time the Brownian motion hits the boundary $\{-1, 1\}$. Prove that there exits some constant $C > 0$ such that

$$\mathbb{P}(\tau > M) \leq Ce^{-M/C}$$

for all $M > 0$.

(6) Let X_n be a sequence of random variables and assume that for any integer $p \geq 1$, we have $\mathbb{E}X_n^p \to m_p$, where m_p is some constant. Assume that $\{m_p\}_{p\geq 1}$ determines a distribution, i.e., there exists a *unique* probability distribution μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$with such moments: $\int_{\mathbb{R}} x^p \mu(dx) = m_p$.

(i) Show that $\{X_n\}_{n\geq 1}$ is tight;

(ii) show that X_n converges in distribution as $n \to \infty$.